

Laughing Waters Harvesting - User & Admin Manual

Laughing Waters Harvesting is a small, self-hosted app that tracks litchis from the moment they're picked in the field, through dispatch to the pack house, to receiving, wages, and reporting. It runs on one local server and is used from several devices at once - a phone or tablet at each field station, one or two devices at the pack house gate, and a computer in the farm office.

This manual is for everyone who touches the app: - **Field capture staff** - picking teams who log crates as they're picked. - **Pack house staff** - whoever receives incoming loads at the gate. - **Farm admin / office staff** - whoever manages workers, wages, reports, and the server itself.

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1. Overview & Concepts

The three device roles

Every device that opens the app is one of exactly three roles:

Role	Used by	What it does
Field	Picking teams, one device per field station	Logs crates as they're picked, sends "picking slips" (loads) to the pack house
Pack House	Gate/receiving staff	Sees incoming loads, checks them in when the truck arrives
Admin	Farm office	Manages workers/teams/blocks/suppliers, calculates wages, runs reports, configures settings

A device only ever shows the screen for its own role - a field tablet never sees the admin screens, and vice versa. This keeps each device simple and hard to misuse by accident.

Why capture is offline-first

Field stations are often out of signal range. The field app is built so that **capturing a crate never depends on having a connection** - every crate is saved on the device first, and quietly synced to the server in the background whenever a connection is available. A picking team can keep working through a signal dropout without losing anything or needing to notice it happened.

Why the oldest-first, color-coded ordering matters

Litchis are highly perishable - quality drops fast once picked, and drops faster again once picked fruit sits waiting in the sun. Two places in the app use the same "traffic light" idea to make that visible at a glance:

- The **pack house's incoming queue** shows the oldest (longest-waiting) load first, colored **green** → **yellow** → **red** as it ages past the thresholds set in Settings (see [chapter 11](#)).
- The **admin Dashboard's Harvesting and In Transit lists** use the exact same coloring, so the office can see at a glance whether fruit is moving through fast enough.

This isn't just a UI nicety - it's the app's main quality-control signal: red means "this has been waiting too long, offload/process it before anything green."

2. Initial Server Setup

The app is a single Python service (FastAPI) that serves its own frontend and stores everything in a local SQLite database file - there's no separate database server or build step to run.

Prerequisites

- Python 3.9
- Network access from every device that needs to reach the app (the same Wi-Fi/LAN the farm already uses, or a Tailscale network if devices need to reach it from outside the farm's own network - the app itself doesn't care which, it just needs to be reachable)

Installation

From the `backend/` folder:

```
python3 -m venv .venv
source .venv/bin/activate
pip install -r requirements.txt
```

Running the server

From the `backend/` folder, with the virtual environment activated:

```
uvicorn main:app --host 0.0.0.0 --port 8000
```

- `--host 0.0.0.0` makes the server reachable from other devices on the network, not just the machine it's running on - this is required for field/pack house devices to connect.
- Pick any free port (8000 is just an example); once running, other devices reach it at `http://<server-ip>:<port>/`.
- Leave this running continuously during harvest season - the app expects to be a long-lived local service, not something started and stopped around each use. (The automatic nightly backup, [chapter 11](#), only fires if the server happens to be running at 02:00 - see that chapter's note on this limitation.)

What happens on first startup

The very first time the server runs, it automatically creates the database and seeds a clean starting baseline:

- Two teams: **Span A** and **Span B** (indunas left blank - fill in via [Master Data](#))
- The farm's **18 real block labels** (7, 8a, 8b, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 22, 23, 34, 35) - variety, tree count, and hectares are left blank for the admin to fill in, since they weren't safe to

guess automatically

- **8 devices:** device-01 through device-05 (field), device-06 and device-07 (pack house), and admin-pc (admin) - see [chapter 3](#) for how these get assigned to physical devices
- A default wage rate of **R3.00/kg**
- One supplier row representing the farm's own fruit ("Laughing Waters (Own)")
- A default admin login: **username admin, password ChangeMe123!**

■■ **Change the default admin password immediately** after first login, via [Settings](#) → [Change admin password](#).

■■ **Do not run `seed_demo.py` on a real farm database**

The repo includes `backend/seed_demo.py`, which fills the database with **fake historical harvest data** for demoing and testing the app. It is **not** part of normal startup and must never be run against a real farm's database - doing so will inject invented workers, lots, and payments alongside real data with no easy way to tell them apart afterward. It's safe to use only against a throwaway/test copy of the app.

3. Device Setup

The very first time any device (phone, tablet, or computer) opens the app's URL, it shows a one-time **Device Setup** screen instead of jumping straight into a role's screen.

1. Enter the **Device ID** you were given by the admin (a dropdown of the preseeded IDs - device-01 through device-07, admin-pc - is offered, but the admin may have added more via [Master Data → Devices](#)).
2. Tap **Continue**.
3. The device looks up that ID's role and **automatically routes itself** to the right screen - Field, Pack House, or Admin - based on what's configured for that device.
4. The device **remembers its ID** after this (stored in the browser, not on the server) - it won't ask again on future visits, and will jump straight to its role's screen.

"Unknown device id"

If the entered ID isn't recognized, the device shows an error and refuses to continue. **Devices are never auto-registered** - an admin must create the device first, in [Master Data → Devices](#), before it can be used. This is intentional: it stops a stray or mistyped device ID from silently attaching itself to the wrong team or role.

Reassigning a device

To point a device at a different role/station (e.g. repurposing a spare tablet), either: - Clear the device's browser data/cache so it forgets its saved ID and shows the setup screen again, or - Simply change that device ID's role/station in [Master Data → Devices](#) - the physical device keeps using the same ID, but the server now treats it differently on its next check-in.

4. Field App - Capturing the Harvest

The field app is what a picking team's device shows once set up. Its job is simple: log every crate as it's weighed, and send a "picking slip" (a load of crates) to the pack house when a truck is ready to take it.

Station header

The top of the screen shows which station/team/induna this device is registered as, and a sync status indicator: **Offline**, **Online - synced**, **Syncing...**, or **Online - sync failed, retrying**.

Identifying the worker: QR scan only

Tap **Scan Worker QR** and point the device's camera at the worker's printed ID badge (see [chapter 5](#) for how badges are made). The app matches the scanned code against known workers and shows the worker's name once matched.

Why QR-only, no typing/dropdown? Picking teams can move dozens of workers through a station in a day - manually finding the right name in a long dropdown list, especially with common surnames, risks misattributing a crate (and therefore wages) to the wrong person. A QR scan is unambiguous.

If the scanned code doesn't match any known worker, the app tells you so and nothing is selected - print or reprint that worker's badge instead of guessing.

Selecting the block

Choose the block being picked from the **Block** dropdown.

Capturing a crate's weight

Use the on-screen numeric keypad to enter the crate's weight in kg, then tap **Save Crate**. The crate is saved immediately to the device - see "Offline-first capture" below - and the running totals update:

- **Crate count** and **total weight** for the current, not-yet-dispatched load
- **Elapsed time**, color-coded green/yellow/red the same way described in [chapter 1](#), so the team can see for themselves when a load has been sitting long enough that it should go.

Offline-first capture

Every crate is written to the device's local storage the instant you tap Save Crate - **capture never waits on a network connection**. In the background, the app checks for a connection every few seconds and quietly syncs anything not yet sent to the server. You do not need to do anything to make this happen, and a signal dropout mid-picking has no effect on capture.

Sending the load: "Send Picking Slip"

When a truck is ready to take the crates picked so far, tap **Send Picking Slip**. You'll be asked for:

- **Crates going now** - defaults to every crate captured so far, but can be reduced if the truck can't take everything (see "Splitting a load" below)
- **Driver name** - required

Once sent, the load moves into "in transit" and will appear in the pack house's queue (see [chapter 6](#)). The screen shows a confirmation banner ("Picking Slip Sent - X crates - Y kg dispatched"), and the load also appears under **Dispatched Lots Today** on the same screen.

Splitting a load

Sometimes a truck arrives before a picking round is finished. If you enter a **Crates going now** value lower than the total captured, the app splits the load: the crates going now are dispatched immediately under the current slip number, and the remaining crates are automatically rolled onto a **new** slip number, ready to combine with whatever gets picked next.

Splitting requires an active connection (the server needs to be the one deciding which crates go on which slip). If offline, the app shows a hint and asks you to either reconnect or just send everything on this load instead of splitting.

The pack house will see both resulting loads flagged as a **split load** with a link back to the other part, so receiving staff know part of this same picking session may arrive separately (see [chapter 6](#)).

5. Worker ID Badges

Badges are what a field device scans to identify a worker (see [chapter 4](#)). An admin generates and prints them from **Master Data** → **Workers**:

- **Print Badges (filtered)** - only the workers currently matching the Farm/Supplier filter
- **Print Badges (selected)** - only the workers with their row checkbox ticked
- **Print Badges (all)** - every active worker

Each badge shows the farm name, the worker's photo (if one's been captured - see [chapter 8](#)), a QR code encoding their employee number, their name, and their employee number printed in large text. Badges print at roughly 9cm × 7cm, several to a page, ready to laminate.

6. Pack House Receiving

This is the screen at the gate where incoming loads are checked in.

The in-transit queue

Every load currently on its way (dispatched from a field station, or logged as an external delivery - see below) appears here, **oldest first**, colored green/yellow/red by how long it's been in transit (the same thresholds as [chapter 1](#), configurable in [Settings](#)). Each card shows the slip number, farm/ supplier, team, driver, crate count, and total kg.

If a load is part of a **split** (see [chapter 4](#)), its card shows a "**Split load - N related slip(s)**" flag. Opening the load shows the related slip(s) with their own status (still in transit, already received, or still being picked) - so receiving staff know to expect (or ask about) the rest of that picking session.

Logging an external delivery

For fruit arriving from another farmer (not via this farm's own field devices), tap **+ Log External Delivery** and fill in:

- **Supplier** - the external farm this fruit belongs to (must already exist in [Master Data → Suppliers](#))
- **Crates**
- **Total Kg**
- **Driver**
- **Notes**

This drops the load straight into the in-transit queue like any other load, ready to be received the same way.

Receiving a load

Tapping a card opens the **Receive** modal, with fields in this order:

1. **Expected crates** - read-only, what was dispatched
2. **Actual crates received** - enter what actually arrived
3. **Condition** - tick any that apply: **Good**, **Damaged**, **Sunburn**, **Wet**, **Other** (more than one can be ticked; all ticked values are recorded together)
4. **Notes**
5. **Received by** - required; the app remembers the last name entered here and prefills it next time, to save retyping for the same gate staff member

Confirming moves the load's status to **received** and records the exact receiving time - this is what feeds the admin Dashboard/Reports' "Received" figures.

7. Admin - Dashboard

The Dashboard is the admin app's landing screen - a farm-wide overview of what's happening right now, all scoped to a shared filter bar at the top.

Filter bar

- **Farm / Supplier** - narrow everything below to one farm/supplier, or leave on "All farms / suppliers"
- **Period start / Period end**, plus quick-fill buttons **Today**, **This Week**, **Season** (season = 1 Jan - 31 Dec of the harvest year set in [Settings](#)) - or set custom dates directly
- **Refresh** - re-fetches everything below using the current filter values

KPI cards

Teams Active, Workers Active, Blocks Active (all counted as "had activity in the selected period", not a static list), Total Kg, Total Crates, Avg Kg/Lot, Avg Kg/Crate, and a breakdown of Harvesting / In Transit / Received crates and kg.

The five collapsible lists

Each starts collapsed, showing its running total in the header; tap to expand for the detail rows.

1. **Harvesting** - loads still being picked (not yet dispatched), oldest-first, color-coded the same as pack house
 2. **In Transit** - dispatched, not yet received, same ordering/coloring
 3. **Received** - newest-received first
 4. **Workers** - per-worker crates/kg/amount-due/avg-kg-per-crate, sorted by kg picked (highest first), showing each worker's **Farm/Supplier**
 5. **Blocks** - per-block crates/kg/avg-kg-per-crate/avg-kg-per-tree
-

8. Admin - Master Data

Master Data has five subtabs for the farm's reference data.

Workers

Employee number, name, ID number, bank/account, WhatsApp number, which farm/supplier they belong to, a photo (captured via the device's camera right in the edit form, or uploaded), and active/inactive. Supports CSV/xlsx **Export** and **Import** for bulk edits, and the [Print Badges](#) buttons.

Teams

ID (e.g. "A"), name (e.g. "Span A"), induna, active.

Blocks

The 18 preseeded block IDs (see [chapter 2](#)) plus name, variety, tree count, hectares, active. Supports CSV/xlsx export/ import, useful for filling in variety/trees/hectares in bulk rather than one block at a time.

Devices

ID, role (field/warehouse/admin), station name, team, induna, data capturer, active. This is where new devices must be added before they can complete [Device Setup](#).

Suppliers

Every farm whose fruit passes through this pack house, including the farm's own row (marked "(Own Farm)"). Contact details and a packing rate (per kg, or per crate if per-kg is left at 0) used for the **Facility Billing** panel on this same subtab - pick a supplier and date range, tap **Calculate**, and see how much facility-use fee that supplier owes for fruit actually received (not still in transit) in that period.

9. Admin - Payments

Calculates and exports wages for a filtered period.

- Same filter bar convention as the Dashboard (Farm/Supplier + Today/This Week/Season/custom dates)
- **Calculate Wages** builds the table below, **grouped by farm/supplier** (own farm first, then alphabetically), each group showing a summary row (worker count, total kg, total wages) followed by that group's workers (name, kg, rate, amount due)
- **Export Wage Sheet** downloads the same grouped breakdown as an `.xlsx` file

This screen only calculates what's **owed** - marking wages as paid, or tracking payment status, is deliberately not handled inside this app; that's managed in whatever external system/process the farm already uses for actually paying workers.

10. Admin - Reports

Downloadable .xlsx reports, all sharing the same filter bar as Payments/ Dashboard (Farm/Supplier + date range):

Report	Contents
Daily Harvest Summary	Crates/kg by block and team for one day
Lot & Receiving Report	Every lot dispatched in the range, with receiving detail once received
Harvesting List	Loads still being picked, matching the Dashboard's Harvesting list
In Transit List	Dispatched, not-yet-received loads
Received List	Received loads
Worker Harvest Report	Per-worker crates/kg/amount-due/avg-kg-per-crate
Block Harvest Report	Per-block crates/kg/avg-kg-per-crate/avg-kg-per-tree

11. Admin - Settings

Data Backup

- **Backup Now** - immediately zips the database and worker photos into a downloadable archive, and adds it to the list below (created date, size, a download link per entry)
- An identical backup also runs **automatically every day at 02:00**, keeping only the **14 most recent** backups (older ones are deleted automatically)

■■ The automatic backup only runs if the server happens to be running at 02:00. If the machine is switched off overnight, that night's backup simply doesn't happen - there's no separate always-on scheduler behind it. Keep the server running continuously during harvest season (see [chapter 2](#)).

Recommended: copy backups off the server regularly. The 14-backup retention only protects against recent mistakes (e.g. accidentally deleting a worker) - it does **not** protect against the server's disk failing entirely, since all 14 copies live on that same machine. Every so often, download the latest backup from this list and copy it somewhere off the machine - a cloud drive such as Google Drive, Dropbox, or similar is a simple, effective option. That off-machine copy is the only real safeguard against losing everything if the hardware fails.

Farm settings

Farm name, location description, current harvest season (year - drives what "Season" means throughout the app), the green→yellow and yellow→red urgency thresholds in minutes (used everywhere the traffic-light coloring appears - [chapter 1](#)), and GPS coordinates (latitude/longitude, or pick a location on the map) - setting these enables automatic weather capture on every dispatched load.

Harvest rate

The per-kg wage rate used by [Payments](#).

Change admin password

Change the admin login password - **do this immediately after first setup** ([chapter 2](#)).

Header weather

The admin screen's header shows a live current-weather readout (temperature, condition, humidity) next to the farm name/clock - a quick at-a-glance check of conditions without leaving the app.

12. Troubleshooting / FAQ

"Unknown device id" on a device's first setup The device ID hasn't been created yet. An admin must add it in [Master Data → Devices](#) first - see [chapter 3](#).

"QR code doesn't match a known worker" when scanning a badge Either the worker doesn't exist in [Master Data → Workers](#) yet, or their badge is stale/misprinted. Reprint the badge from Workers after confirming the worker record exists (see [chapter 5](#)).

Field app shows "Online - sync failed, retrying" The device has a connection but the server rejected or couldn't be reached for the last sync attempt. Nothing is lost - captured crates stay queued on the device and the app keeps retrying automatically every few seconds.

"Reconnect to send a partial load, or dispatch everything now" when sending a picking slip You tried to split a load (send fewer crates than captured) while offline. Splitting needs a connection because the server decides the split; either wait for a connection or send the full load instead.

Forgotten admin password There's currently no self-service "forgot password" flow in the app - a new password can only be set from inside Settings while already logged in. If the admin password is lost entirely, restoring access requires direct access to the server's database rather than anything available from the app's screens.

Annexe A: Data Field Reference

Field-by-field reference for every stored record type, with a realistic example value and any limitation worth knowing. Types/defaults/foreign keys below match `backend/models.py` exactly.

Worker

Field	Type	Example	Notes / Limitations
id	text	"001"	Primary key - the employee number. Must be unique and must match the number printed on the worker's badge.
first_name	text	"Sipho"	
last_name	text	"Dlamini"	
id_number	text	"8501015800083"	Free text, not validated.
bank	text	"Nedbank"	
account	text	"9963334018"	

whatsapp_number	text	"+27821234567"	Optional, not currently used to send anything automatically.
supplier_id	number (optional)	2	Which farm this worker belongs to. Left blank for the farm's own workers.
photo_filename	text	"001.jpg"	Set automatically when a photo is captured/uploaded - not hand-edited.
active	true/false	true	Inactive workers are hidden from most pickers/drop but kept for history.

Team

Field	Type	Example	Notes / Limitations
id	text	"A"	Primary key. Short code, e.g. a single letter.
name	text	"Span A"	
induna	text	"Samuel Mthembu"	
active	true/false	true	

Block

Field	Type	Example	Notes / Limitations
id	text	"8a"	Primary key - a real farm block label. One of the 18 preseeded labels (see chapter 2); not free-form.
name	text	"Block 8a"	
variety	text	"Mauritius"	

trees	number	450	Whole number of trees on the block.
hectares	decimal	2.3	
active	true/false	true	

Device

Field	Type	Example	Notes / Limitations
id	text	"device-01"	Primary key - must be entered exactly on the physical device's setup screen (chapter 3).
role	enum	"field"	Must be exactly one of field, packhouse admin.
station	text	"Field Station 1"	

team_id	text (optional)	"A"	Which team this field device belongs to. Not applicable to pack house/admin devices.
induna	text	"Samuel Mthembu"	
data_capturer	text	""	Free text, optional.
active	true/false	true	
last_seen	timestamp	(auto)	Updated automatically every time the device checks in - not editable.

Supplier

Field	Type	Example	Notes / Limitations
id	number	1	Primary key, auto-assigne
name	text	"Jansen Boerdery"	

contact_name / contact_phone / contact_email	text	"Piet Jansen" / "082-555-1234" / —	All optional.
is_own_farm	true/false	false	Exactly one supplier row in the whole system should ever have this set to true - that row represents the farm's own fruit.
packing_rate_per_kg	decimal	1.50	If greater than 0, this rate is used for Facility Billing; otherwise the per-crate rate below is used.

packing_rate_per_crate	decimal	25.00	Only used when the per-kg rate is 0.
active	true/false	true	

HarvestRecord (one crate)

Field	Type	Example	Notes / Limitations
uuid	text	(auto-generated)	Primary key, generated on the capturing device - lets a retry after a dropped connection safely resubmit the same crate without duplicating it.
timestamp	timestamp	2026-07-08T09:14:00Z	When the crate was captured.
worker_id	text (optional)	"001"	Set via the QR badge scan (chapter 4) - required in practice even though nullable in the schema.
block_id	text (optional)	"8a"	Required in practice.
weight_kg	decimal	12.4	A single crate's weight - typically in the 8-20kg range for litchi crates.
deduction_kg	decimal	0.0	Exists in the schema for a future "afrekkings" (waste/reject deduction) workflow, but there is currently no screen to enter a non-zero value - always 0 in this version of the app.

lot_id	number (optional)	21	Always set at capture time, pointing at a placeholder load that becomes a real dispatch once "Send Picking Slip" is used.
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Lot (a picking slip / load)

Field	Type	Example	Notes / Limitations
slip_number	text	"device-01-20260708091400"	Unique. Auto-generated from the device ID and a timestamp; a split creates a second, related slip number.
timestamp	timestamp	2026-07-08T09:14:00Z	Dispatch time - the basis for the urgency color-coding
supplier_id	number	1	Which farm this load's fruit belongs to.

driver	text	"Sello"	Required when dispatching.
total_crates / total_kg	number / decimal	18 / 238.5	
status	enum	"in_transit"	One of created (still being picked), in_transit (dispatched), received, processing
received_at	timestamp (optional)	—	Set the moment pack house staff confirm receipt.
weather_temp / weather_humidity / weather_condition	decimal / decimal / text	24.1 / 55 / "Clear"	Captured automatically at dispatch time, only if GPS coordinates are set in Settings.

split_from_slip_number	text (optional)	—	Only set on the "leftover" slip created by a split (chapter 4) - points back at the original slip it was carved out of.
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ReceivingRecord

Field	Type	Example	Notes / Limitations
lot_id	number	21	Which load this receiving entry belongs to.
expected_crates / actual_crates	number	18 / 17	

discrepancy	number	-1	Calculated automatically as actual - expected - not entered directly.
condition	text	"Good, Sunburn"	Free text, but in practice a comma-joined list of whichever of Good/Damage were ticked.
received_by	text	"Elsa"	Required - the app remembers and prefills the last value entered on this device.

Payment

Field	Type	Example	Notes / Limitations
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worker_id	text	"001"	
period_start / period_end	date	2026-07-01 / 2026-07-31	
total_kg	decimal	318.6	Sum of that worker's net crate weights in the period.
rate_applied	decimal	3.00	The per-kg rate in effect when calculated.
amount_due	decimal	955.80	Calculated - not directly editable. There is intentionally no "paid" flag or status field on this record (chapter 9); payment status is tracked outside this app.

RateSetting

Field	Type	Example	Notes / Limitations
effective_date	date	2026-07-01	
rate_type	enum	"per_kg"	per_kg or per_crate
default_rate_per_kg	decimal	3.00	Used when rate_type is per_kg.
tier_rates_json	text (JSON)	{"1": 2.5, "1.5": 3.5, "2": 4.5}	Only relevant if using per_crate - maps a crate-size tier to its own rate.

SystemSetting

Field	Type	Example	Notes / Limitations
farm_name	text	"Laughing Waters (Bekfontein)"	

farm_location	text	"Bekfontein, Mpumalanga"	Free text description, not used for weather (see gps_lat/gps_lon)
current_harvest_year	number	2026	Drives what "Season" means throughout the app (chapters 7, 9, 10).
green_to_yellow_minutes / yellow_to_red_minutes	number	90 / 150	The urgency color thresholds referenced throughout chapters 1, 4, 6, 7.
gps_lat / gps_lon	decimal (optional)	-25.572747 / 31.606722	Setting both enables automatic weather capture on dispatch.

AdminUser

Field	Type	Example	Notes / Limitations
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username	text	"admin"	
password_hash	text	(hashed)	Never shown or exported anywhere - only a hash is stored.